DIGITAL TWIN CHALLENGE 2026

Prototype & Code Submission

GlucoTwin

A Personalized Glucose Digital Twin

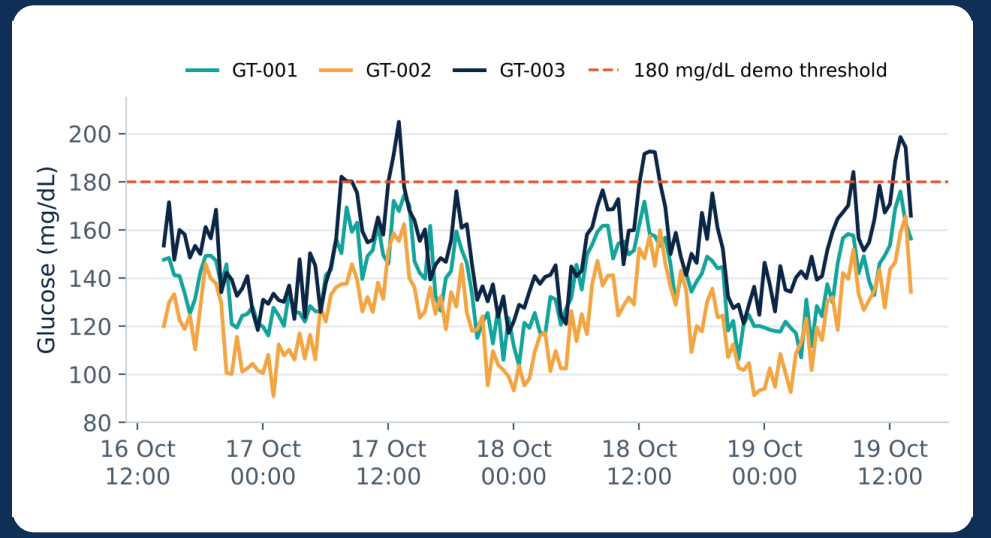
Research prototype for glucose trend forecasting and event analysis using synthetic data.

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Individual project

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SYNTHETIC GLUCOSE TRACES • 3 PATIENTS



Forecast MAE 8.373 vs 9.278 persistence baseline (mg/dL)
402 held-out rows • Random Forest • +30 min horizon

RESEARCH PROTOTYPE ONLY • NOT CLINICALLY VALIDATED

SECTION 1

Problem Statement

A technical problem framed around time-dependent glucose patterns

1 Time-dependent patterns

Glucose readings form patterns over time that are difficult to interpret from isolated values.

2 Short-term forecasting

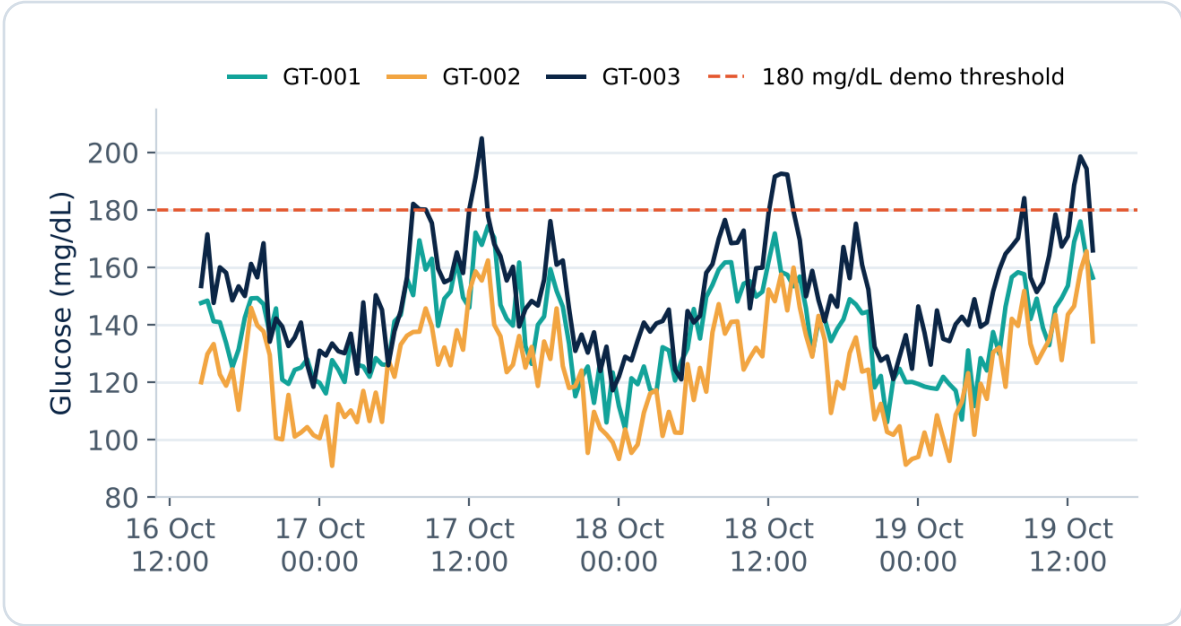
Forecasting offers a way to explore how a digital representation of glucose measurements can be analyzed.

3 Event analysis needs care

Potentially significant events require evaluation with attention to false positives and false negatives.

4 Scope of this study

Tasks are studied on synthetic data only; the project does not demonstrate clinical benefit.



Synthetic glucose traces (last 3 days) for GT-001, GT-002, GT-003

Different baselines and variability per patient; the dashed line marks the 180 mg/dL demonstration event threshold used in the event model.

SECTION 2

Objectives & Healthcare Use Case

Exploratory software workflow — not a clinical tool

01 Generate synthetic patient profiles and glucose time-series records.

02 Explore short-term glucose forecasting and compare predictions with a persistence baseline.

03 Evaluate event classification at selected thresholds using classification metrics.

04 Visualize history, forecasts, events, data records and model summaries through Streamlit.

POTENTIAL USE CASE

Research and education for testing data → model → visualization workflows on safe, synthetic records.

BOUNDARY

No diagnosis, treatment, medication decisions, or clinical deployment are supported by this project.

Educational research software — not a medical device.

SECTION 3

Proposed Solution

A reproducible pipeline from synthetic data to interactive inspection

DATA

Synthetic patient & glucose records

3 patients • 14 days • 30-min readings

→

MODELS

Forecasting & event classification

Random Forest regressor / classifier

→

EVALUATION

Metrics, thresholds, baseline

Persistence baseline • P / R / F1

→

DASHBOARD

Interactive visualizations

Streamlit + Plotly data explorer

DATASET AT A GLANCE

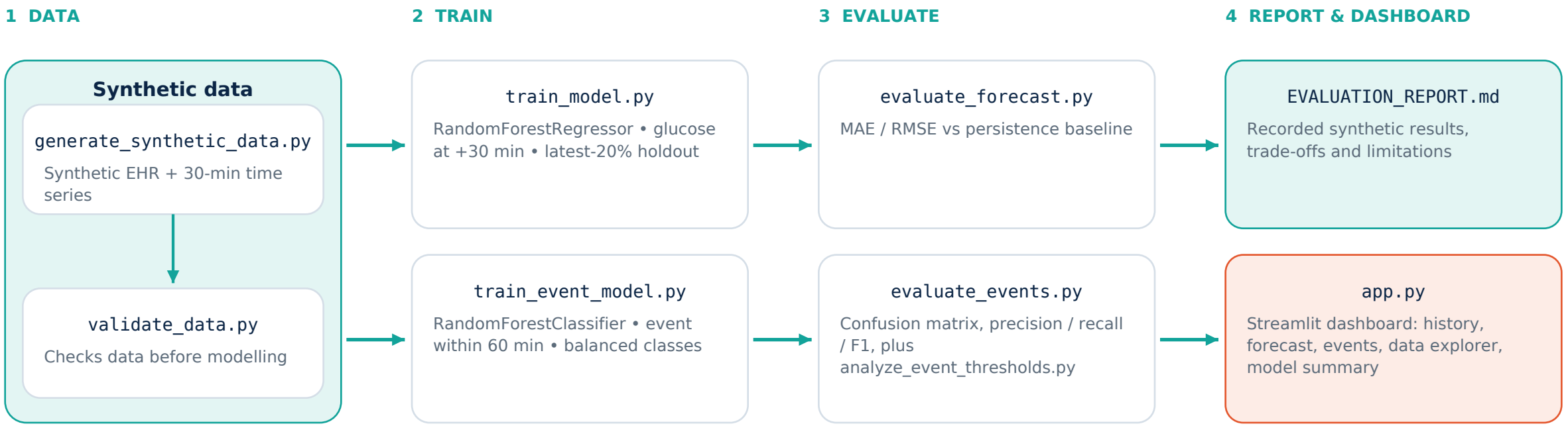
3	14 days	30 min	2,016	402
synthetic patients	of glucose records	sampling interval	glucose time-series rows	held-out evaluation rows

Every record is tagged SYNTHETIC_DEMO. No real patient data is used anywhere in this pipeline.

SECTION 4

System Architecture

Repository components and their roles (file names as in the GitHub repository)



Data flow generate → validate → train (2 models) → evaluate → report + dashboard

Both models use the same 10 input features and the same latest-20%-per-patient chronological holdout.

SECTION 5

Technical Stack & AI/ML Workflows

Tools and configuration taken from the repository's training scripts

STACK

Python | Pandas | NumPy | Scikit-learn | Plotly | Streamlit

FORECASTING WORKFLOW

Model	RandomForestRegressor — 150 trees, max depth 8, min leaf 4
Target	Glucose 30 minutes ahead (1 step)
Baseline	Persistence: next value = current value
Metrics	MAE, RMSE (mg/dL)

EVENT-CLASSIFICATION WORKFLOW

Model	RandomForestClassifier — same settings, balanced class weights
Label	Glucose ≥ 180 mg/dL at +30 or +60 min (demo definition)
Threshold	0.5 default; 0.1–0.9 explored
Metrics	Precision, recall, specificity, F1, Brier

Shared design: 10 input features — current glucose, 3 lags, heart rate, 30-min steps, sleep proxy, age, diabetes duration, HbA1c. Latest 20% of each patient's timeline is held out; event labels use future readings and are never model inputs.

SECTION 6

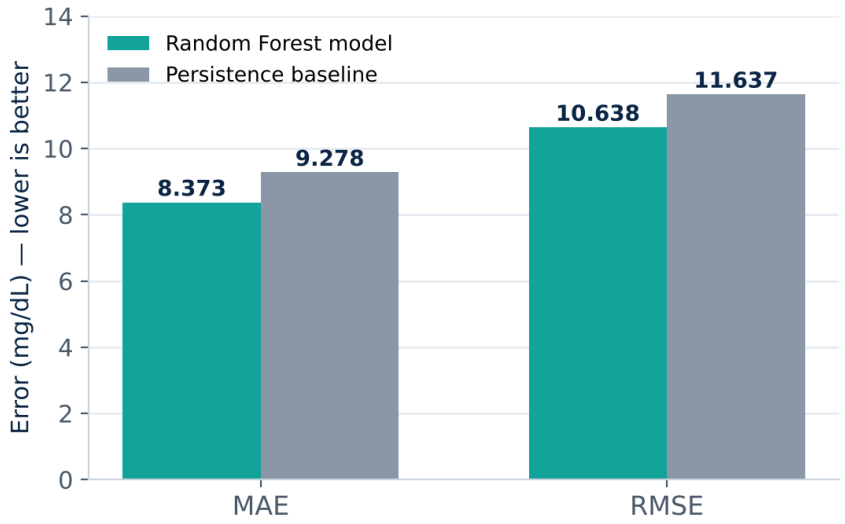
Evaluation Results — Forecasting

Recorded from the synthetic evaluation (EVALUATION_REPORT.md) and reproduced from data/forecast_evaluation.csv

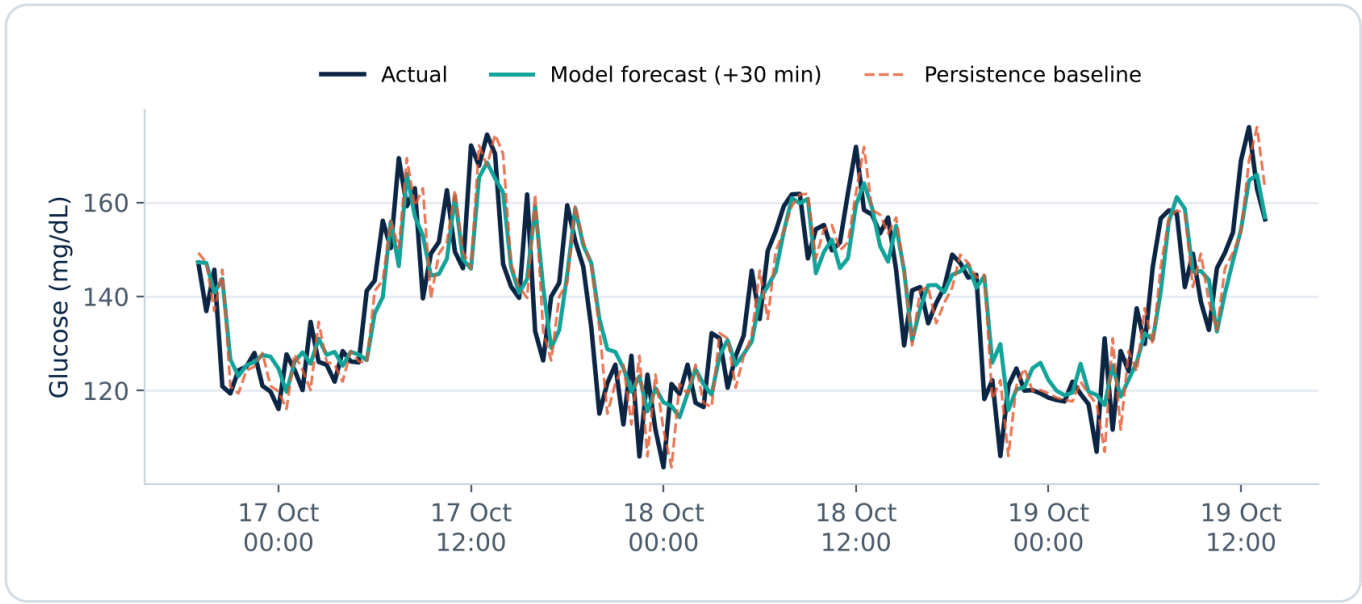
FORECAST MAE (mg/dL)
8.373
Persistence baseline: 9.278 (−9.75%)

FORECAST RMSE (mg/dL)
10.638
Persistence baseline: 11.637 (−8.59%)

EVALUATED ROWS
402
Latest 20% per patient • none skipped



Error vs persistence baseline



Held-out forecast trace for GT-001 (actual vs model vs baseline, +30 min)

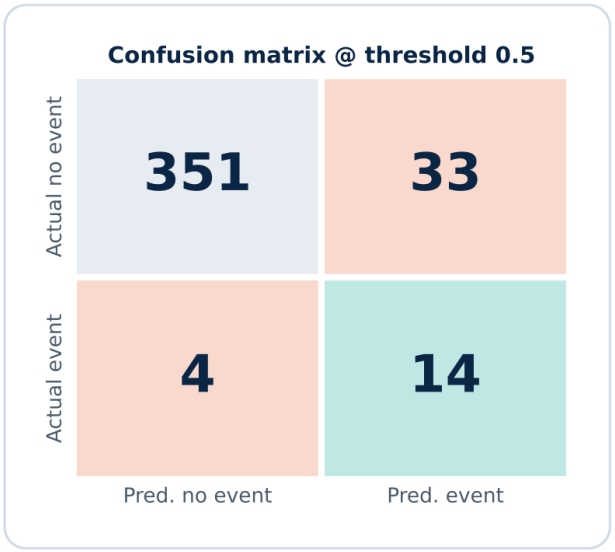
Model error is lower than the persistence baseline on this synthetic evaluation. This does not establish clinical accuracy or generalization.

SECTION 6

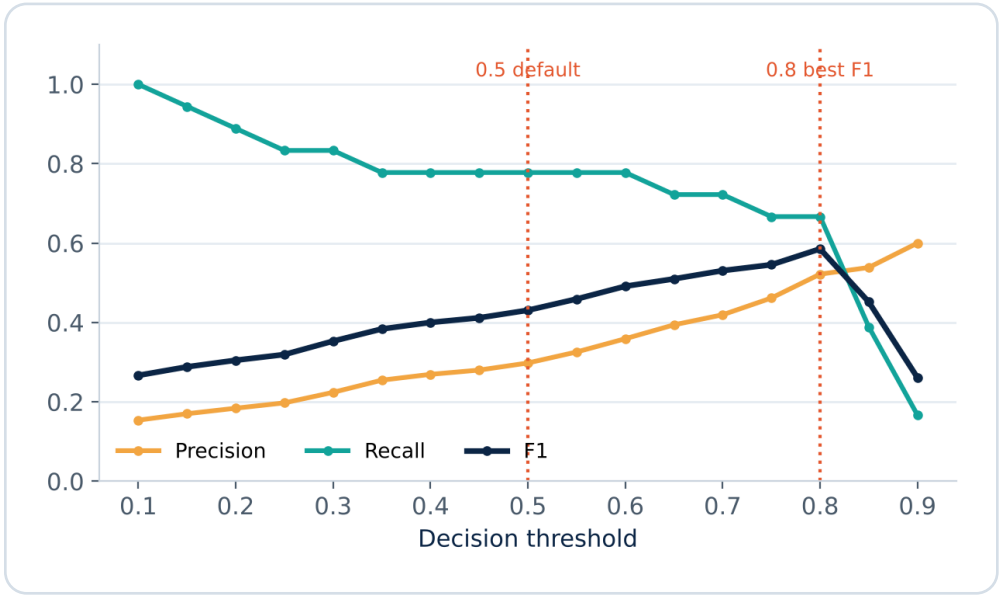
Evaluation Results — Event Classification

Threshold-dependent behaviour on 402 held-out rows (18 events, 384 non-events)

0.4308	0.2979	0.7778	0.9141	0.0619
EVENT F1 @ 0.5	PRECISION	RECALL	SPECIFICITY	BRIER SCORE



Confusion matrix (default threshold)



Precision, recall and F1 across thresholds

0.5 vs 0.8 threshold

	0.5	0.8
Precision	0.298	**0.522**
Recall	0.778	**0.667**
F1	0.431	**0.585**
False pos.	33	**11**
False neg.	4	**6**

Threshold 0.8 gives the best F1 (0.585) with fewer false positives but more false negatives — a trade-off, not an operational recommendation.

Research prototype only • Synthetic data • Not clinically validated

SECTION 7

Prototype Walkthrough

Interactive Streamlit dashboard with six tabs (app.py)

1 Glucose history

Interactive time-series view of the synthetic patient records.

2 Forecast

Timeline, actual-vs-predicted and error-distribution views against the baseline.

3 Events

Event predictions, probabilities and threshold-dependent outcomes.

4 Data explorer

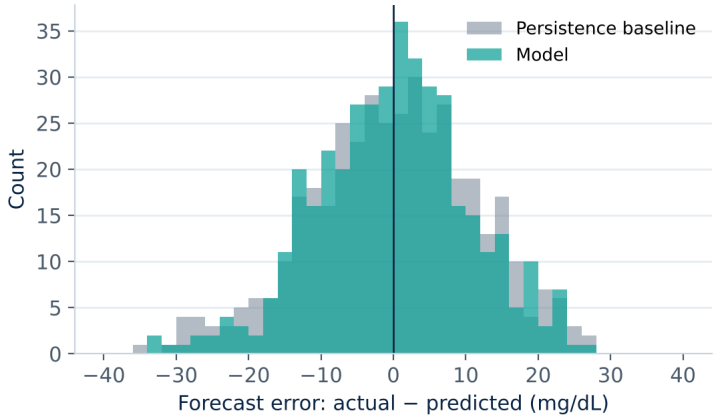
Browse and download synthetic patient and glucose records.

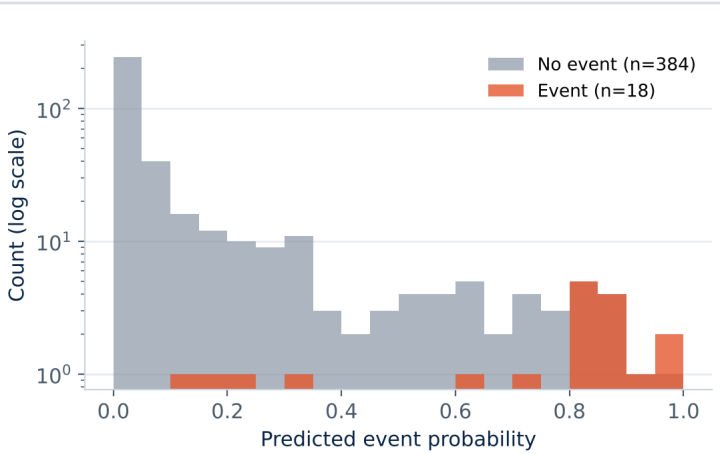
5 Model summary

Metrics and visual summaries of both models.

6 About / disclaimer

Scope, synthetic-data notice and safety boundary.





Events tab: predicted probability by true class

Figures are rendered from the repository's evaluation CSVs (data/forecast_evaluation.csv, data/event_evaluation.csv) with the same data the dashboard reads; they illustrate the tab content and are not screen captures.

SECTION 8

Limitations & Future Work

Responsible interpretation of a healthcare-related prototype

LIMITATIONS

- Synthetic data only; no clinical validation has been performed.
- Results depend on dataset construction, evaluation design, model choices and classification thresholds.
- Only 3 synthetic patients and 18 positive events in the test split — estimates are noisy.
- False positives and false negatives remain important limitations (precision 0.298 at 0.5).

FUTURE WORK

1. **Real-world data**
   Appropriately licensed datasets
2. **Temporal validation**
   Stronger, independent splits
3. **Calibration**
   Probability calibration assessment
4. **Model comparison**
   Additional model families
5. **External evaluation**
   With clinical oversight

GlucoTwin is educational research software, not a medical device.

SECTION 9

Repository & Demonstration Plan

Final presentation and 15–20 minute demo checklist

PUBLIC GITHUB REPOSITORY

github.com/EDDY05ee/GlucoTwin

https://github.com/EDDY05ee/GlucoTwin

QUICK START

```
$ pip install -r requirements.txt
$ python data/generate_synthetic_data.py
$ python train_model.py && python train_event_model.py
$ streamlit run app.py
```

DEMO VIDEO (YouTube, unlisted)

ADD LINK AFTER RECORDING AND UPLOADING

Also add the final link to the repository README.

DEMO CHECKLIST (suggested timing ≈ 18 min)

1	**Introduce** Problem, objectives and synthetic-data safety boundary	2 min
2	**Repository** Show setup commands and main files	3 min
3	**Dashboard** History, forecasts, events, data explorer, model summary	7 min
4	**Evaluation** Open EVALUATION_REPORT.md; explain metrics and limitations	4 min
5	**Close** Future work and the safety disclaimer	2 min